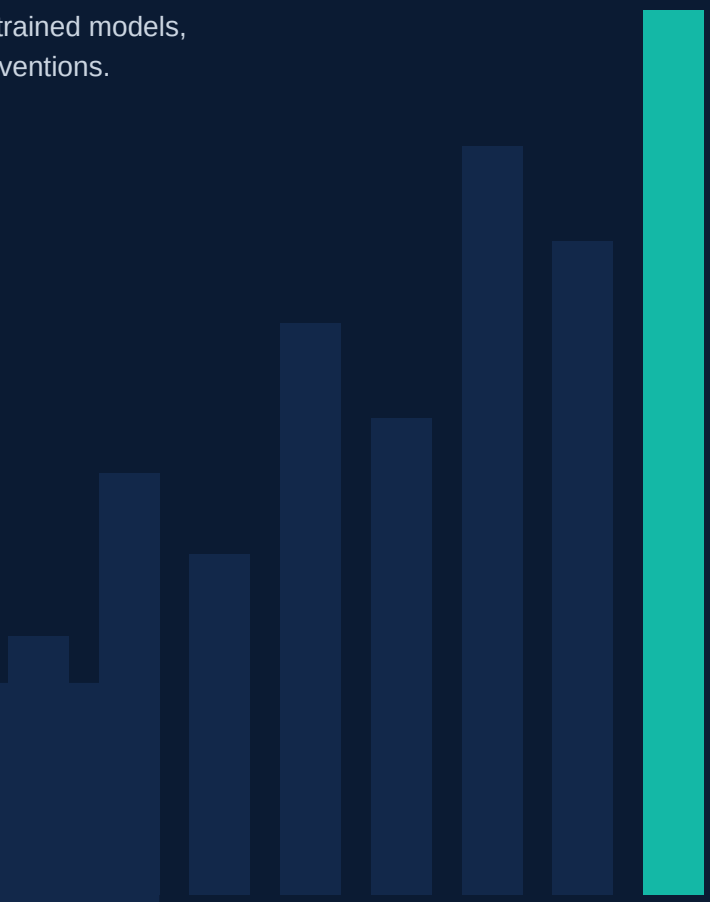
PROJECT DOCUMENTATION · USER GUIDE

CredX

Early Warning System for Credit Risk Monitoring

Detect borrower stress before default: repayment behaviour, exposure and cash flow, scored by rules and trained models, with human-in-the-loop investigation and interventions.



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FastAPI · Astro · XGBoost · scikit-learn

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Who this guide is for. Credit analysts, relationship officers, credit and portfolio managers who use CredX day to day, and developers or evaluators who need to install, run and integrate it.

SECTION 01

Introduction

CredX is an **Early Warning System (EWS)** for lending portfolios. It watches repayment behaviour, credit exposure and customer cash flow, and flags borrowers who show signs of financial stress *before* they default, so a lender can act while there is still time to help the customer and protect the exposure.

The problem

Traditional monitoring reacts to days-past-due (DPD) thresholds. By the time an account is 90 DPD, most of the options are gone. Early signals exist much sooner: small but repeated payment delays, rising card utilisation, falling monthly inflows, new loans taken to service old ones. Individually they look minor; together they tell a story. CredX reads those signals together.

What CredX does

- **Ingests** customers, loans, repayments, exposure snapshots and cash-flow records from CSV or Excel files.
- **Scores** every customer with four transparent business rules and three trained machine-learning models, producing a 0 to 100 warning score and a Low / Medium / High / Critical band.
- **Explains** each score in plain language: which signals drove it and by how much.
- **Tracks trends** month by month, so analysts can see whether a customer is improving or deteriorating.
- **Prioritises** the watchlist by risk and exposure, so the biggest problems get attention first.
- **Supports action**: investigation cases, recommended interventions, notes, information requests and a second-person approval for any customer-impacting step.
- **Investigates with AI agents**: specialist agents gather evidence, form and challenge hypotheses, and hand every conclusion to a human reviewer.

> **Design principle:** CredX advises, people decide. No endpoint changes a loan, a credit limit or contacts a customer. Every intervention is a recorded human action.

SECTION 02

System Architecture

CredX is a monorepo with three parts that work together:

Layer	Technology	Responsibility
Frontend	Astro, TypeScript	Dashboard, watchlist, customer detail, portfolio monitor, priority queue, interventions, workspace and upload pages. Talks to the API through `/api`.
Backend API	FastAPI (Python 3.14), SQLAlchemy, Alembic	Authentication (JWT access + refresh tokens), data ingestion, assessment, monitoring, cases, interventions, agent investigations, RBAC and audit.
Scoring engine	core/ package: rules + models	Four business rules and three trained models (XGBoost and scikit-learn gradient boosting) combined into one warning score.
AI layer	LLM gateway (Nebius, DeepSeek)	Plain-language explanations, case briefings and agent reasoning. Optional: if disabled or unavailable, deterministic scoring keeps working.
Data store	SQLite	Customers, facilities, repayments, exposure, cash flow, assessments, risk history, cases, interventions, investigations and an append-only audit trail.

Request flow

- The analyst uploads data on the **Upload** page; the API validates each row and stores it.
- **Run assessment** sends every customer through the scoring engine and records an assessment.
- The monitoring service replays the pipeline at each month end to build a **risk history**.
- Pages read the watchlist, trends, priority queue and workspace from the API and render them.
- Actions (cases, notes, classifications, interventions) are written back with the acting user recorded.

Repository layout

```
CredX/
  Frontend/           Astro web app (pages, components, lib)
  backend/
    app/api/v1/       auth, users, ews, monitoring, investigations,
                      reviews, admin, system
    app/api/v2/       role-governed, agent-backed EWS endpoints
    app/services/     scoring, risk history, prioritisation, ...
    app/agents/       director, specialists, challenger, adjudicator
    migrations/       Alembic schema migrations
    sample_data_v3/   test portfolio (30 customers, 12 months)
  core/
    rules/            delays, missing history, exposure, temporary
    models/           cash-flow, warning-score, deterioration models
```

SECTION 03

Getting Started

Prerequisites

- Python 3.14 (with `uv` or `pip`)
- Node.js 20+ and npm
- Optional: a Nebius API key for AI explanations and agent reasoning

1. Start the backend

```
cd backend
uv sync --locked
uv run python scripts/init_env.py        # creates backend/.env with a random JWT key
uv run alembic upgrade head
uv run uvicorn app.main:create_app --factory --reload --port 8000
```

Without uv, create a virtual environment and install the hash-pinned requirements:

```
python3.14 -m venv .venv && . .venv/bin/activate
python -m pip install --require-hashes -r requirements-dev.txt
```

Interactive API documentation is then available at `http://127.0.0.1:8000/docs` (Swagger UI) and `/redoc`.

2. Start the frontend

```
cd Frontend
npm install
npm run dev                              # http://localhost:4321
```

The development server proxies `/api` to the backend on port 8000.

3. Optional configuration (backend/.env)

Setting	Purpose
`NEBIUS_API_KEY`	Enables AI explanations, case briefings and agent reasoning.
`LLM_ENABLED`	Set to false to run fully deterministic (no LLM calls).
`BOOTSTRAP_ADMIN_EMAIL`	Promotes this registered user to risk administrator at startup, while no administrator exists.
`CORS_ORIGINS`	Allowed browser origins when the frontend is hosted separately.
`DATABASE_URL`	Defaults to SQLite at backend/data/credx.db.

First login: open the app, choose **Register**, create an account and sign in. The landing page leads to the dashboard.

SECTION 04

Using the Application

The left navigation gives access to every area. Each page is described below in the order an analyst normally uses them.

Page	What you do there
Upload	Import data files. Choose the data type (customers, loans, repayments, exposure, cashflow), pick a CSV or Excel file and upload. Rows that fail validation are listed with the reason; re-uploading the same file is safe (idempotent).
Dashboard	The portfolio at a glance: total debtors, how many were assessed, the split across Low / Medium / High / Critical, and the customers needing attention.
Watchlist	Every assessed customer with warning score, risk band and key signals. Filter by band and open any customer.
Customer detail	One customer: profile, loans, repayment history, exposure, cash flow, assessment history and the plain-language explanation of the score. Alerts can be classified here.
Portfolio Monitor	Run the assessment for the whole portfolio, rebuild the monthly risk history, see the trend of each band over time and which customers moved.
Priority	The work queue: customers ranked by risk *and* exposure into priority tiers, so a large exposure at moderate risk can outrank a small one at the same probability.
Workspace	The investigation desk for one customer: correlated signals, trend, recommendations, an AI case briefing, cases, notes, information requests and event classification.
Interventions	All proposed and active interventions, with owner, priority, target date and status. Customer-impacting interventions wait for approval by a second signed-in user.

Typical daily flow

Step	Action
1	Upload the latest repayment, exposure and cash-flow files.
2	Run the assessment from the Portfolio Monitor.
3	Open the Priority queue and work from the top tier down.
4	In the Workspace, review signals, read the briefing, open a case and add notes.
5	Correct data issues: classify a late payment as a processing error so it is excluded and the score recalculates.
6	Propose an intervention; a second user approves it if it affects the customer.
7	Track progress on the Interventions page and close the case with an outcome.

SECTION 05

How Risk is Scored

Every customer receives a **warning score from 0 to 100**. The score combines transparent business rules, which analysts can reason about, with trained models that capture patterns the rules miss.

Business rules (core/rules)

Rule	What it measures
Repayment delays	Frequency and severity of late payments (days past due) across instalments.
Missing history	Gaps and missed instalments; how complete the repayment record is.
Increasing exposure	Rising outstanding balances, credit-limit utilisation and number of active loans.
Temporary vs sustained	Separates a one-off slip from a sustained pattern, so a single late payment is not treated like chronic stress.

Trained models (core/models)

Model	Algorithm	Training data	Test AUC
Cash-flow deterioration	scikit-learn Gradient Boosting	12-month cash flows for 307,511 Home Credit applicants	0.888
Warning score	XGBoost (hist)	Home Credit applications, instalments, bureau and card balances (307,511 customers)	0.678
Repayment deterioration	Gradient Boosting	Home Credit instalment and card history (62,712 customers)	0.773

Customers with less than three months of data are tagged **thin-file**: the models that need history are skipped and the rules carry the score, so new customers are not penalised for missing data.

Risk bands

Band	Score	Meaning
Low	0 to 19	Healthy. Routine monitoring.
Medium	20 to 44	Early signs. Monitor and verify.
High	45 to 69	Clear stress. Investigate and plan an intervention.
Critical	70 to 100	Severe. Immediate action to protect the exposure.

Explainability. Each assessment lists its contributing signals and the model contributors. The Explain action turns them into a short narrative in plain language; every number in that narrative is checked against the computed figures.

SECTION 06

Monitoring & Interventions

A single score is a snapshot. CredX also keeps the **history**, so analysts see direction as well as level.

Risk trends

The pipeline is replayed at every month end to build a risk snapshot per customer per month. The trend view shows the score path and labels the customer *improving*, *stable* or *deteriorating*.

Signal correlation

Moderate signals that occur together (for example falling income, rising card use and short delays) raise the warning level more than any of them alone would.

Prioritisation

Customers are ranked into priority tiers by combining risk with exposure. Two customers with the same default probability are ordered by how much is at stake.

Event classification

An analyst can classify a repayment event, such as a processing error or auto-debit failure. Classified events can be excluded from scoring; the score is recalculated and both before and after values are kept.

Cases

A case groups the work on one customer: owner, status, notes, information requests and the final disposition, including a corrected category with a reason.

Interventions

Recommended actions are generated from the scenario (for example monitor and verify, restructure, protect exposure). Each has an owner, priority and target date. Customer-impacting interventions require approval from a second signed-in user.

SECTION 07

Agent Investigations

For complex cases CredX can open an **investigation** carried out by a team of AI agents, with people in control at every decision point.

Agent	Role
Director	Plans the investigation and assigns tasks to specialists.
Specialists	Analyse repayment, exposure and cash-flow evidence with deterministic tools and record findings with citations.
Challenger	Questions each hypothesis and looks for evidence that contradicts it.
Adjudicator	Weighs findings and challenges and proposes a classification for human review.

Human in the loop

- Analysts add context, challenge hypotheses, review findings and validate evidence.
- Managers make the classification, intervention and portfolio decisions; approvals respect each manager's delegated approval limit.
- Investigations can be escalated, reassigned, reopened or cancelled, and monitoring instructions can reopen them automatically when conditions are met.
- Agent outputs, human decisions and audit entries are **append-only**: they cannot be edited or deleted.
- Budgets cap LLM usage per investigation; cost and token usage are visible.

Roles

Role	Typical permissions
Relationship officer	Views assigned customers, adds context, requests investigations.
Credit analyst	Uploads data, runs assessments, investigates, reviews findings.
Credit manager	Decides classifications and interventions within an approval limit.
Portfolio manager	Portfolio-level decisions and intelligence.
Risk administrator	Assigns roles and scopes, manages versioned institution policy.
Auditor	Read-only access to investigations and the audit log.

New accounts start **unassigned** and have no access to governed endpoints until a risk administrator assigns a role.

SECTION 08

Sample Data Walkthrough

The repository includes a made-up test portfolio in `backend/sample_data_v3`: 30 customers (T3-001 to T3-030) over 12 months, with a monthly exposure snapshot per facility.

Load it

- Open **Upload** and upload, in this order: customers.csv, loans.csv, repayments.csv, exposure.csv, cashflow.csv.
- Open the **Portfolio Monitor** and click **Run assessment**. Everyone is scored and the monthly risk history is built.

Scenarios to try

Customer	Scenario	What to look for
T3-001	Stable prime home loan, always on time	Low and stable; large exposure falls under "protect exposure"
T3-002	30-45 DPD early in the year, on time for 7 months	Trend labelled improving
T3-003	Income -24%, card use 40% to 82%, short delays	Signals correlate and raise the level; ranks above T3-004
T3-004	Same signals on a small loan	Same probability, lower priority (small exposure)
T3-005	One instalment 6 days late	Temporary miss: monitor and verify
T3-006	One instalment 31 days late (auto-debit failure)	Classify as processing error: score drops from 39.5 to about 13

The full list of scenarios is in `sample_data_v3/scenarios.csv`. Earlier test sets are in `sample_data` and `sample_data_v2`.

SECTION 09

API Reference

All endpoints are served under `/api`. The complete, always-current reference is the OpenAPI document at `/docs` and `backend/docs/API.md`.

Area	Endpoints
Health	GET /health
Authentication	POST /api/v1/auth/register, /login, /refresh, /logout; GET /api/v1/users/me
EWS core	POST /api/v1/ews/upload, /assess; GET /ews/watchlist, /ews/customer/{id}, /ews/portfolio; POST /ews/alert/{assessment_id}, /ews/explain/{customer_id}
Monitoring	POST /api/v1/ews/monitoring/history/rebuild; GET /ews/monitoring/customer/{id}/trend, /monitoring/priority, /monitoring/dashboard
Workspace	GET /api/v1/ews/workspace/{id}; POST /ews/workspace/{id}/briefing
Cases	POST, GET /api/v1/ews/cases; PATCH /cases/{id}; POST /cases/{id}/notes, /info-requests, /classification; PATCH /info-requests/{id}
Event classification	POST, DELETE /api/v1/ews/repayments/{id}/classification
Interventions	POST, GET /api/v1/ews/interventions; PATCH /interventions/{id}; POST /interventions/{id}/approve
Investigations	POST, GET /api/v1/investigations; GET /{id}, /{id}/events; POST notes, tasks, challenge, review, validate-evidence, evidence-requests, assign, escalate, complete-review, decisions, reopen, cancel
Reviews	GET /api/v1/reviews/queue
Administration	GET /api/v1/admin/users; PUT /admin/users/{id}/role; POST, DELETE /admin/users/{id}/scopes; GET, POST /admin/policies
System	GET /api/v1/system/status, /audit/logs, /portfolio/intelligence
Governed EWS (v2)	The role-checked, agent-backed EWS endpoints under /api/v2/ews

Authenticated endpoints expect `Authorization: Bearer <access token>`. Errors use a consistent JSON body with status 401 (unauthenticated), 403 (forbidden), 409 (conflict) or 422 (validation).

SECTION 10

Security, Governance & Team

- Passwords are hashed; logins are rate-limited and accounts lock after repeated failures.
- Short-lived JWT access tokens with rotating refresh tokens; logout revokes the session.
- Role-based access control with per-customer scopes for governed endpoints.
- Uploaded identifiers and labels containing markup are rejected.
- Append-only records and a full audit log of every allowed and denied action.
- AI is advisory: deterministic scoring works without it, and AI narratives are checked against computed numbers.

Scope note. CredX is a decision-support system and a development build, not a core banking system. It holds no account balances and performs no transfers, payments or customer contact.

The Team

CredX was designed and built by **Porotta Pythoners**.

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